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Kim et al.

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(54) **DEVICE CHIP SCALE PACKAGE INCLUDING A PROTECTIVE LAYER AND METHOD OF MANUFACTURING A DEVICE CHIP SCALE PACKAGE**

(58) **Field of Classification Search**

CPC . H01L 21/563; H01L 21/565; H01L 23/5283; H01L 23/3114; H01L 23/3128;

(Continued)

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(63) Continuation of application No. 17/720,211, filed on Apr. 13, 2022, now Pat. No. 11,990,411, which is a (Continued)

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(51) **Int. Cl.**

H01L 23/528 (2006.01)

H01L 21/56 (2006.01)

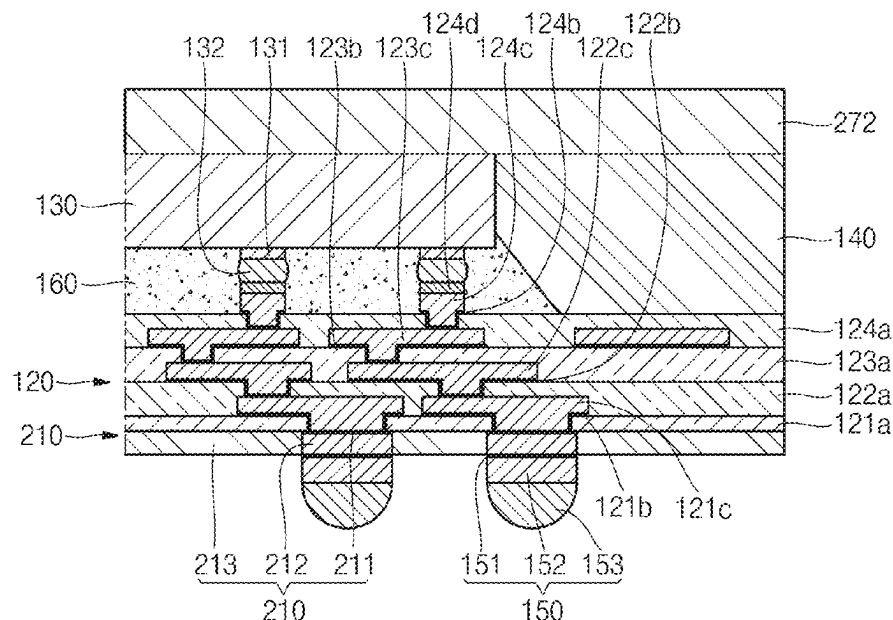
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ABSTRACT

In one example, a semiconductor device comprises a redistribution layer (RDL) substrate having a top surface and a bottom surface, wherein the RDL substrate comprises a filler-free dielectric material, an electronic device on the top surface of the RDL substrate, an electrical interconnect on the bottom surface of the RDL substrate and electrically coupled to the electronic device, a first protective material contacting a side surface of the electronic device and the top

(Continued)



surface of the RDL substrate, and a second protective material contacting a side surface of the electrical interconnect and the bottom surface of the RDL substrate. Other examples and related methods are also disclosed herein.

20 Claims, 11 Drawing Sheets

Related U.S. Application Data

continuation of application No. 16/387,924, filed on Apr. 18, 2019, now abandoned.

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- (52) **U.S. Cl.**
 CPC **H01L 23/3114** (2013.01); **H01L 23/3128** (2013.01); **H01L 24/09** (2013.01); **H01L 24/17** (2013.01); **H01L 2224/0231** (2013.01); **H01L 2224/02373** (2013.01); **H01L 2224/02381** (2013.01); **H01L 2224/0401** (2013.01)
- (58) **Field of Classification Search**
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 USPC 257/690
 See application file for complete search history.

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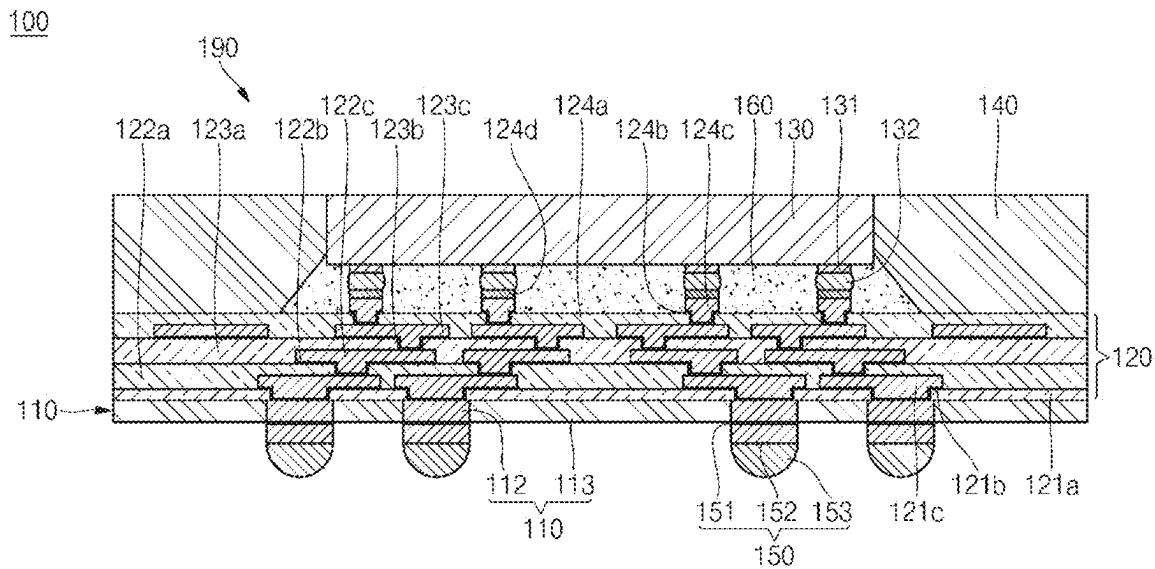


FIG. 1

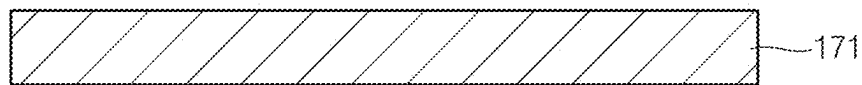


FIG. 2A

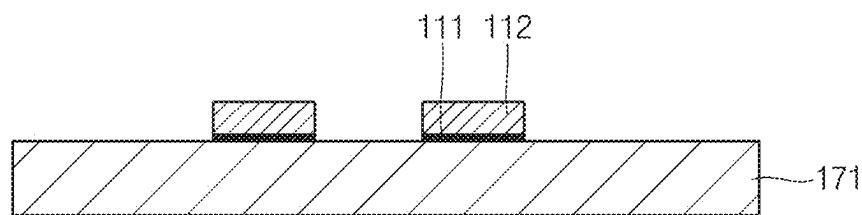


FIG. 2B

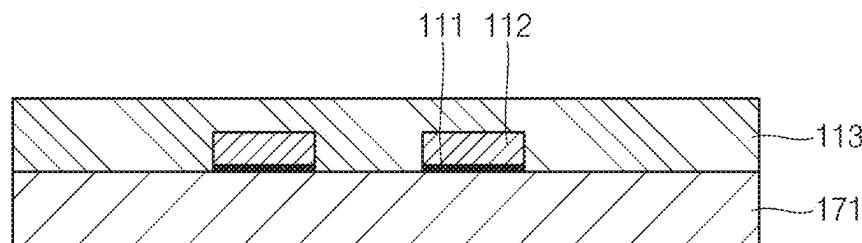


FIG. 2C

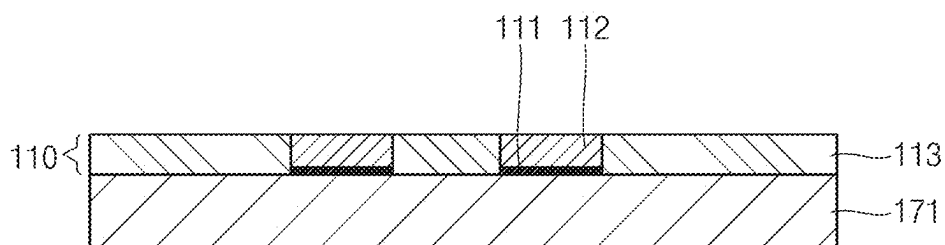


FIG. 2D

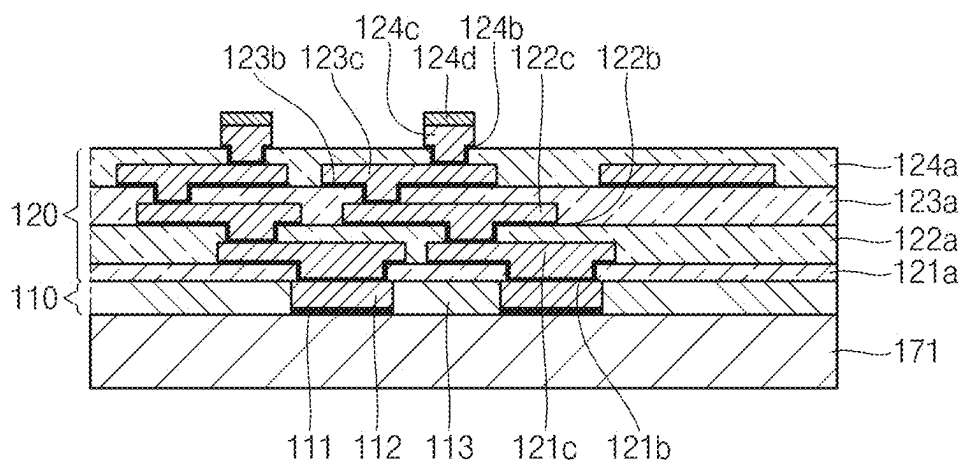


FIG. 2E

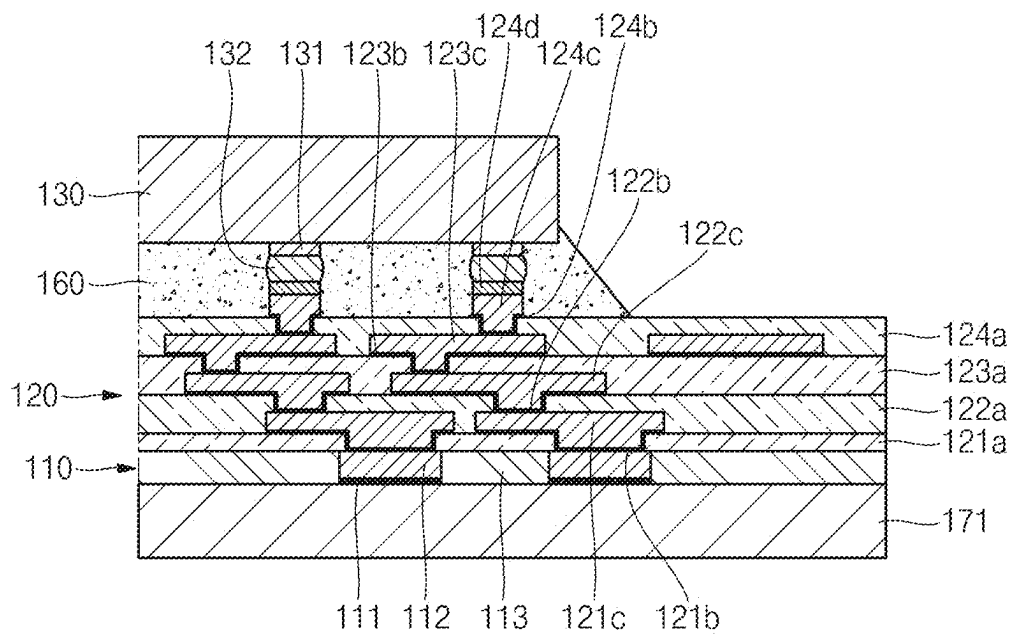


FIG. 2F

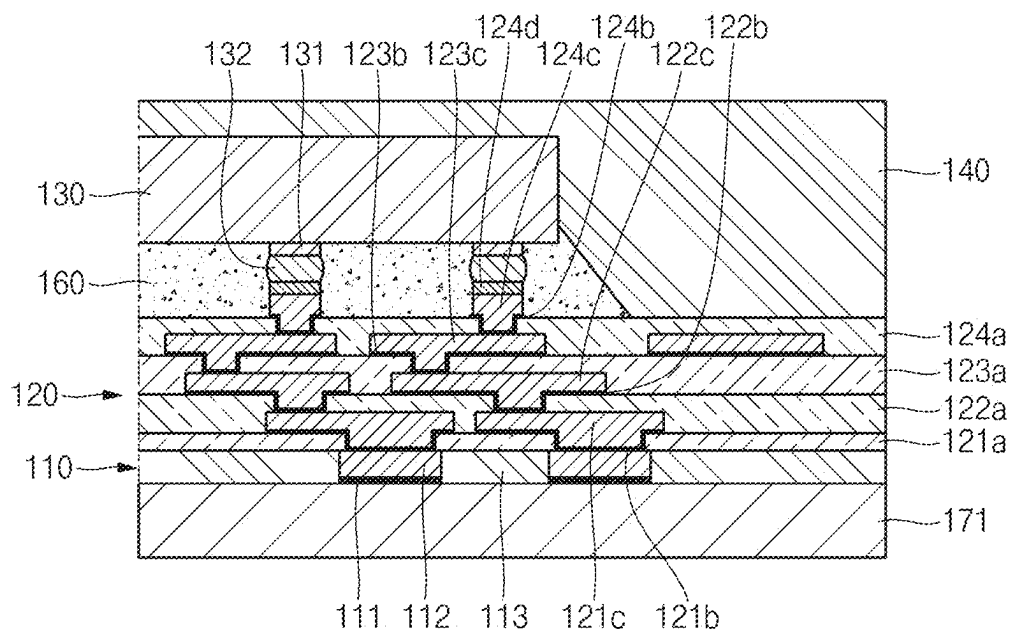


FIG. 2G

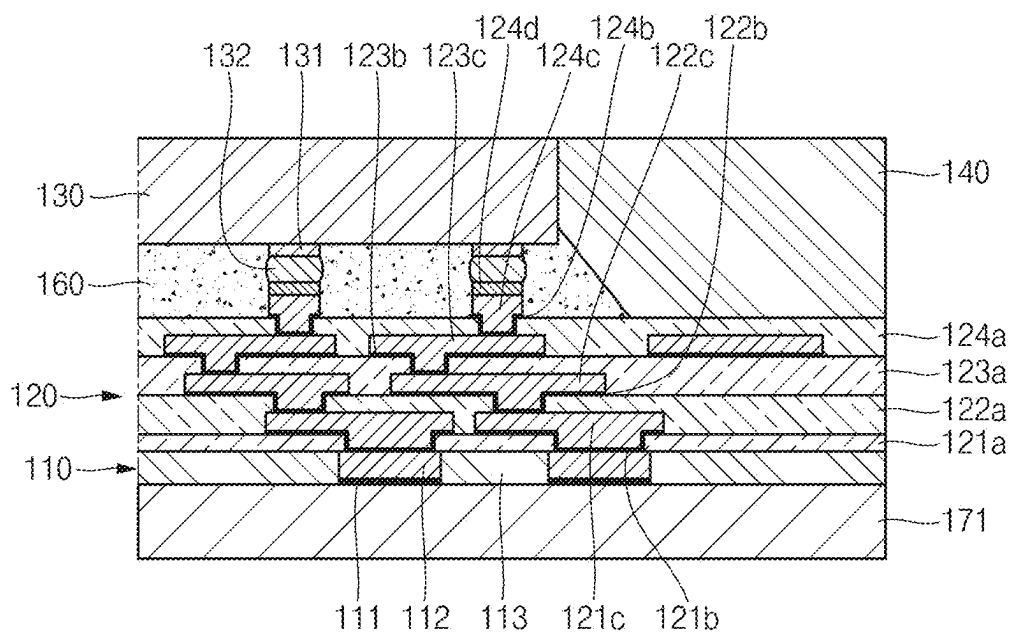


FIG. 2H

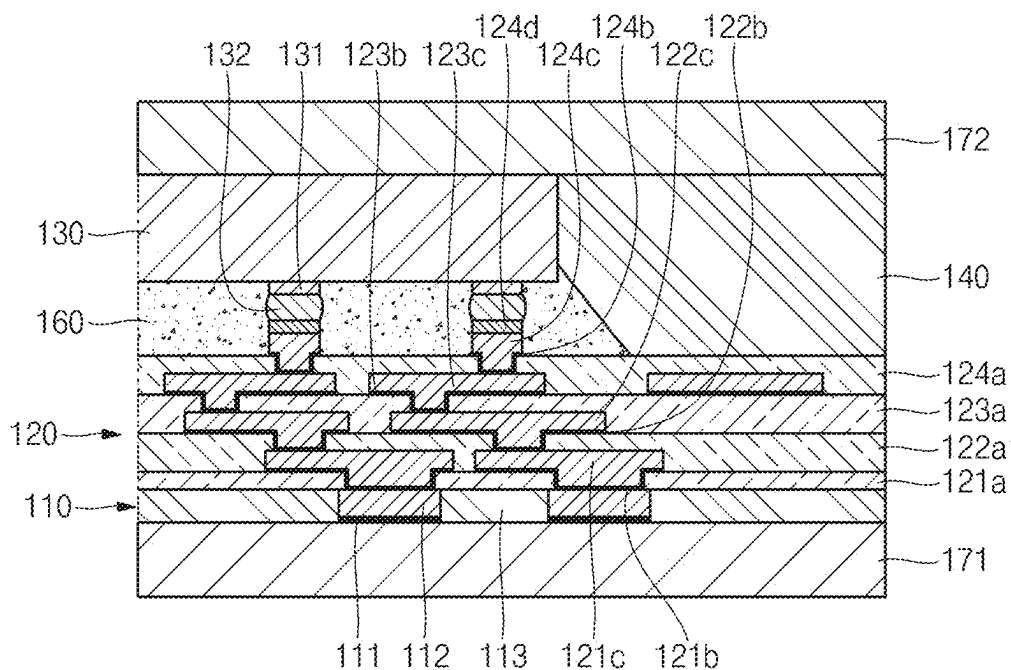


FIG. 2I

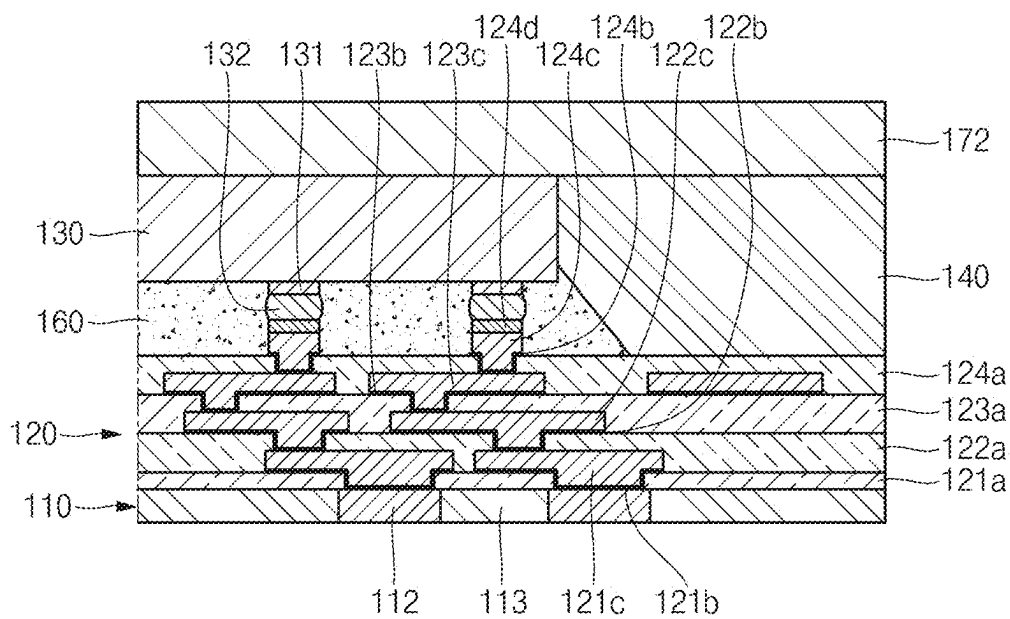


FIG. 2J

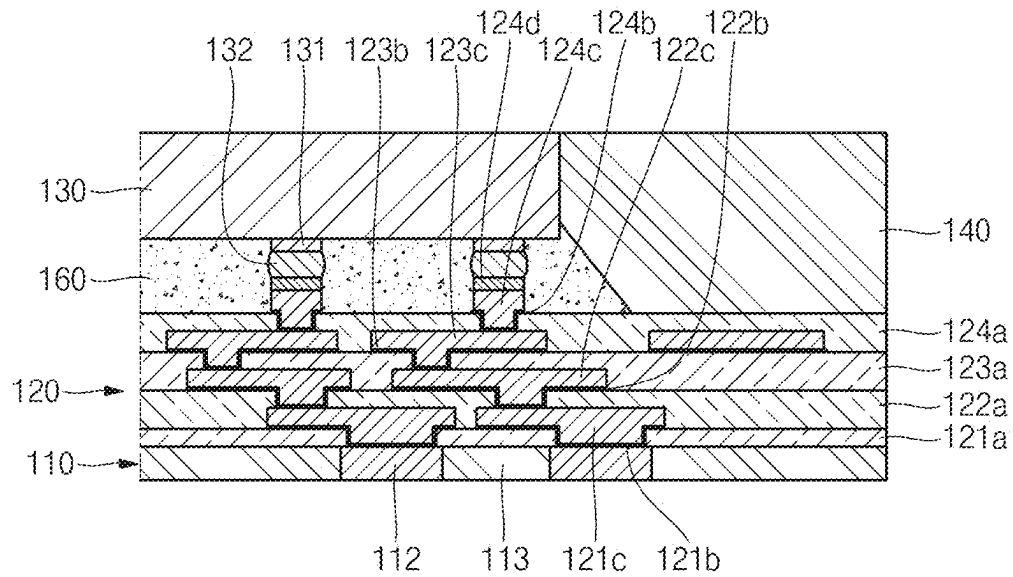


FIG. 2K

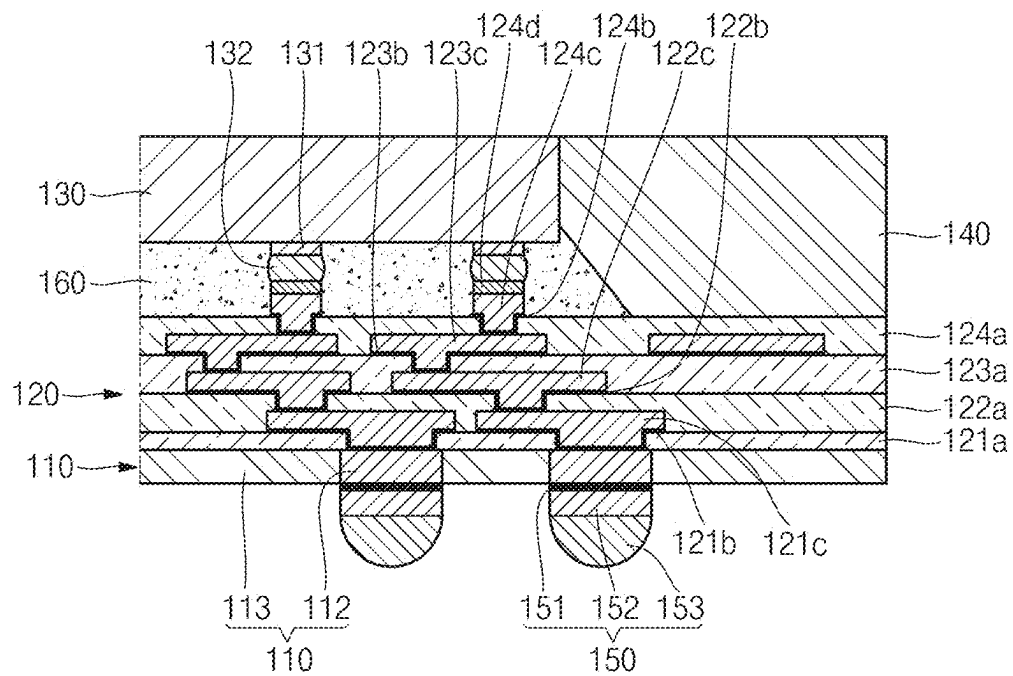


FIG. 2L

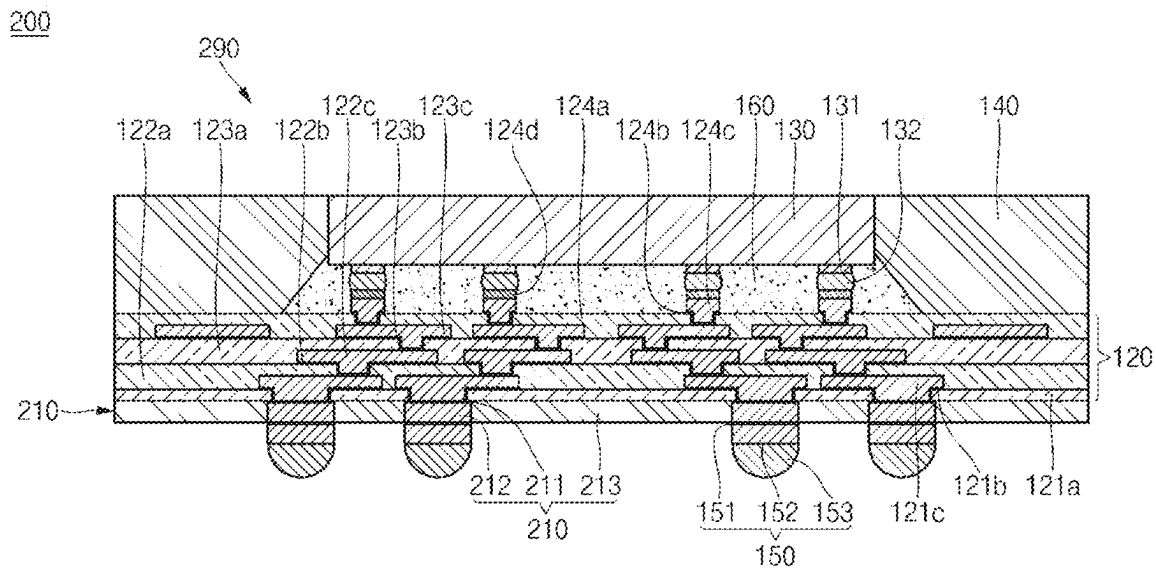


FIG. 3

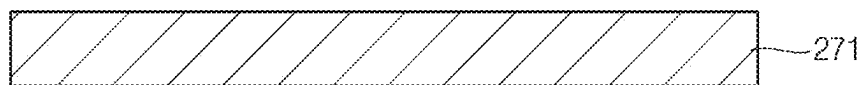


FIG. 4A

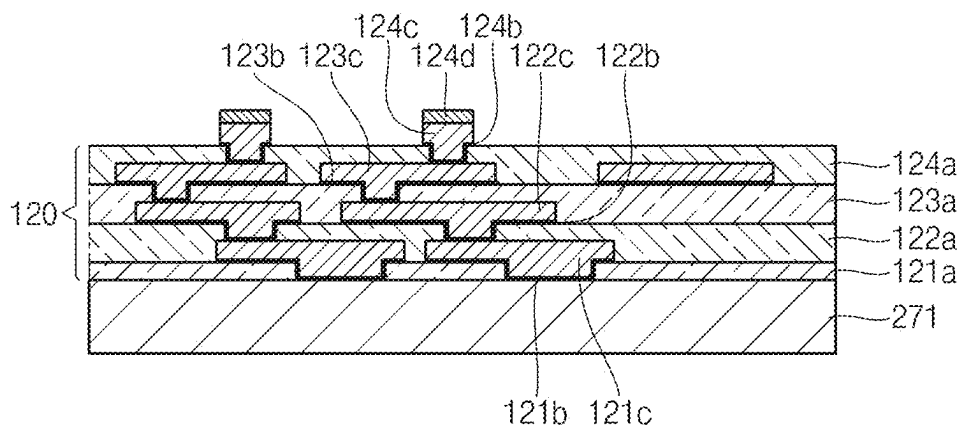


FIG. 4B

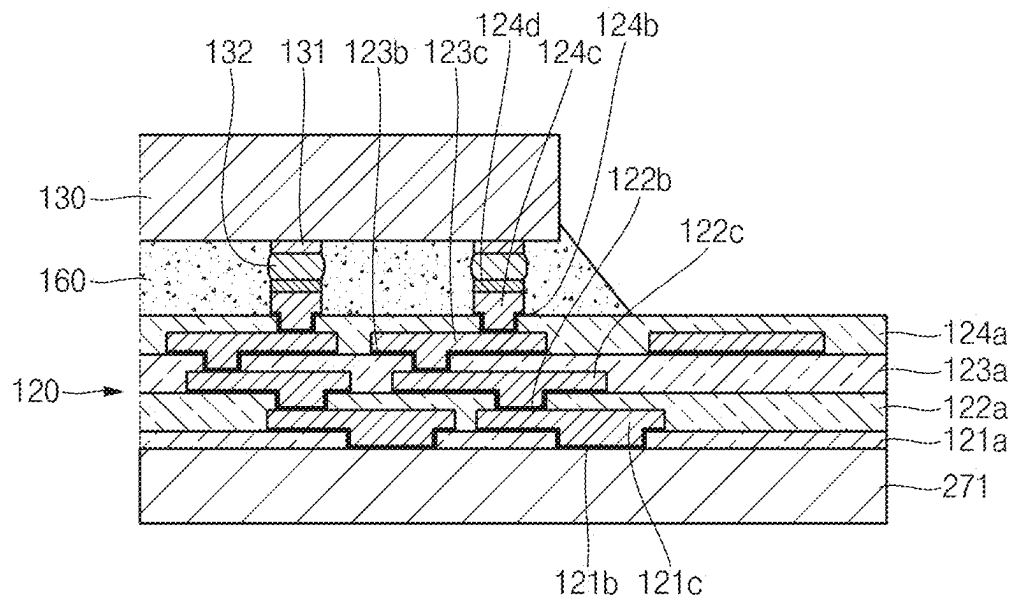


FIG. 4C

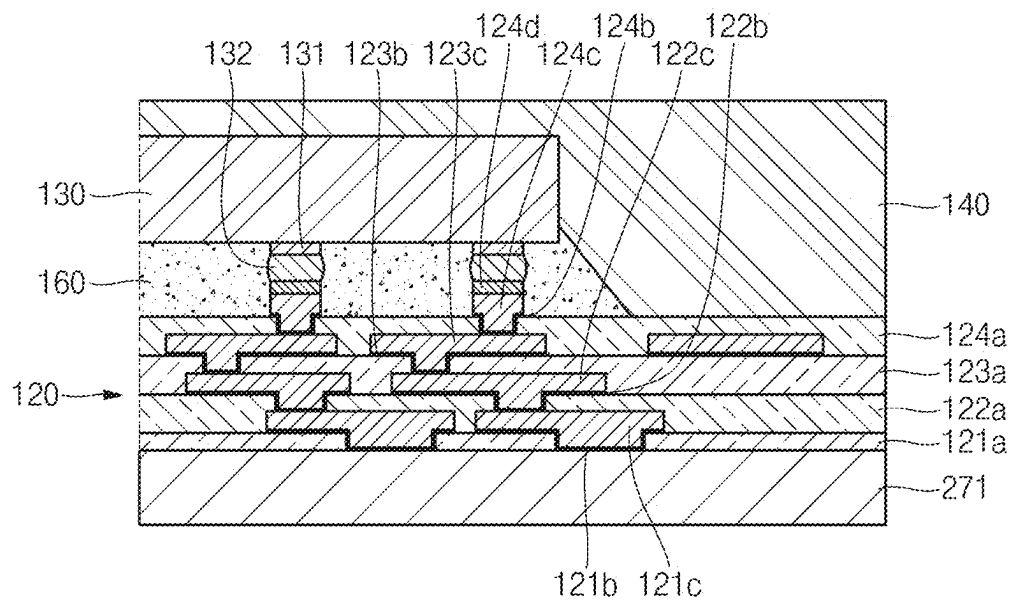


FIG. 4D

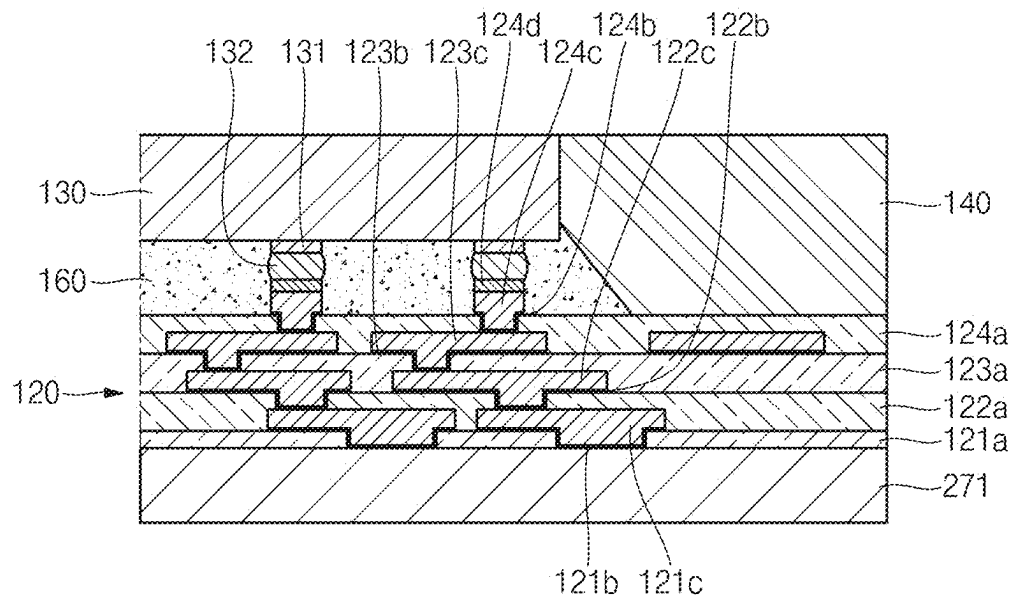


FIG. 4E

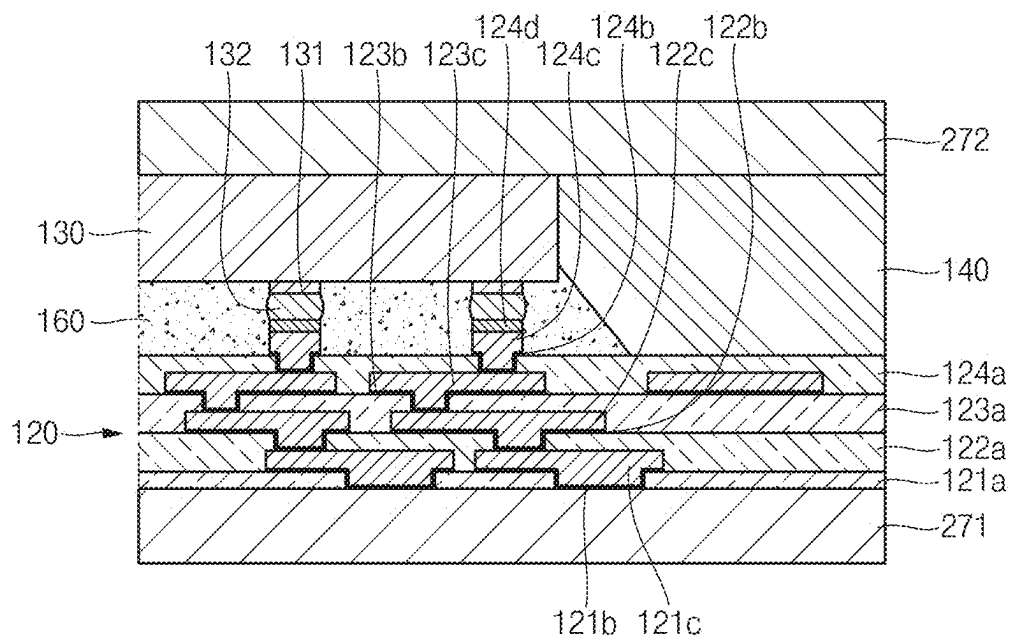


FIG. 4F

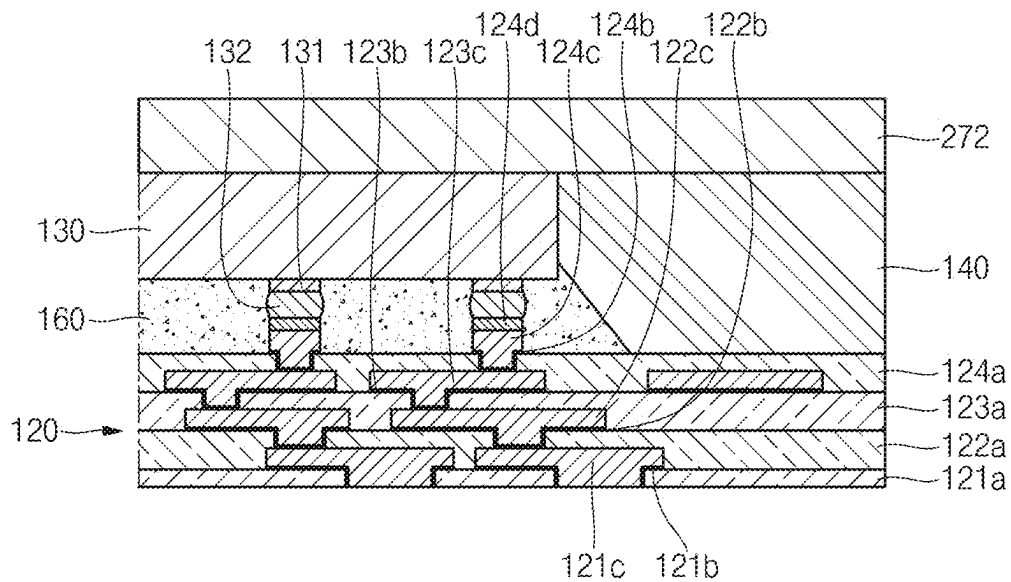


FIG. 4G

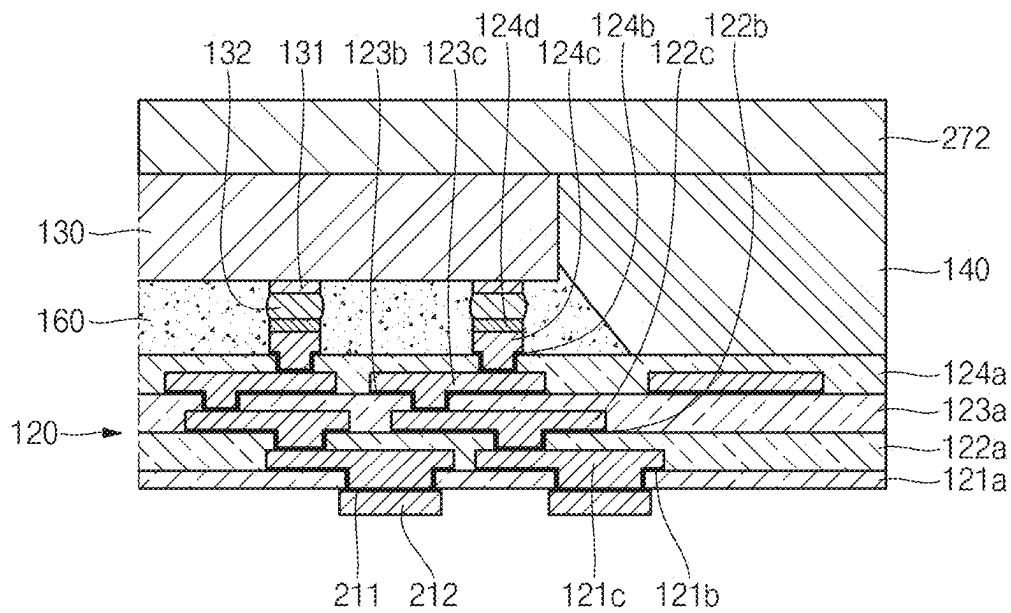


FIG. 4H

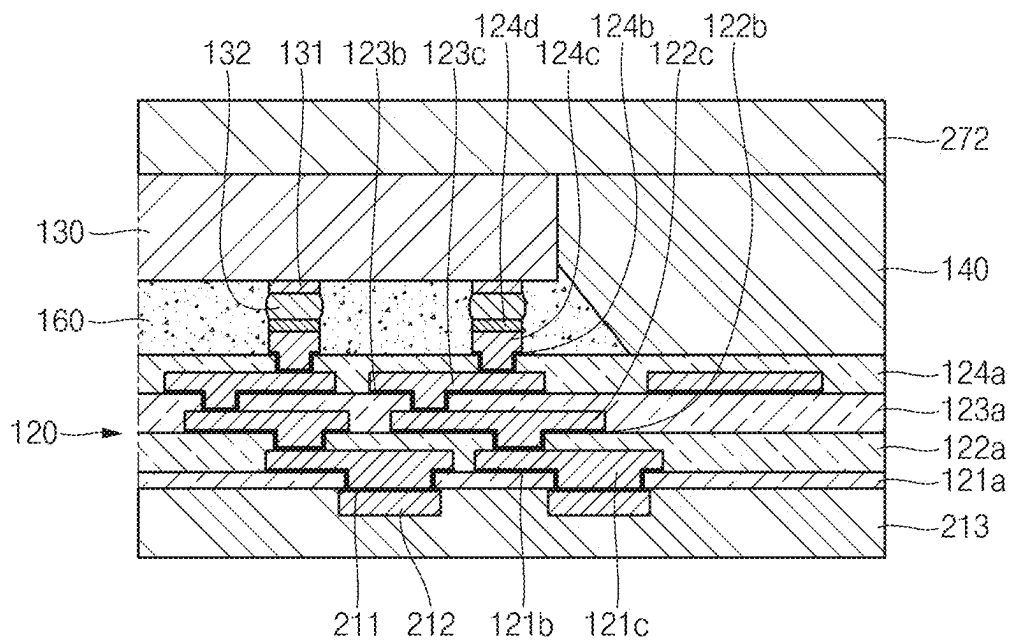


FIG. 4I

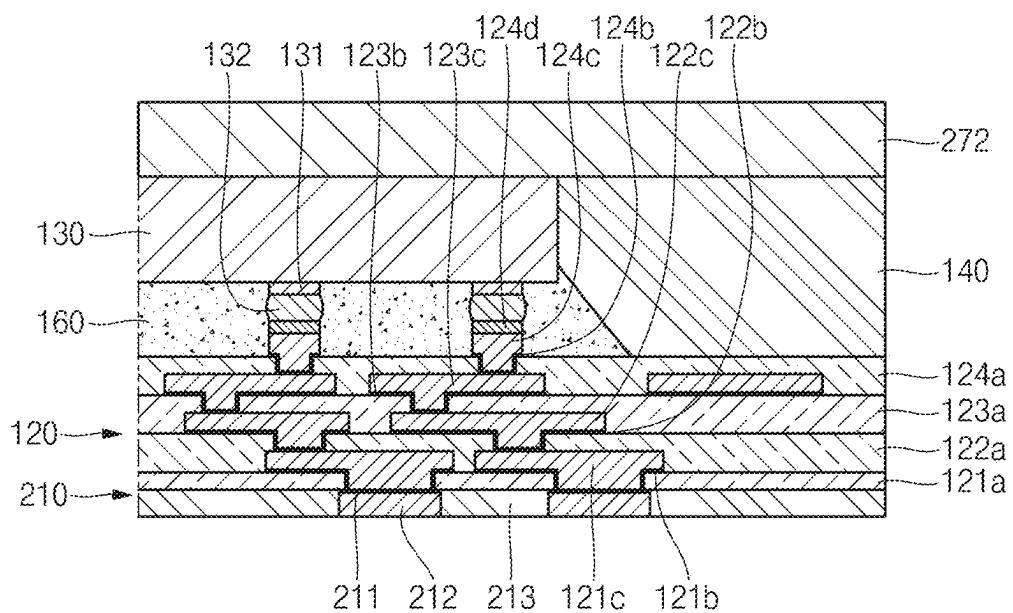


FIG. 4J

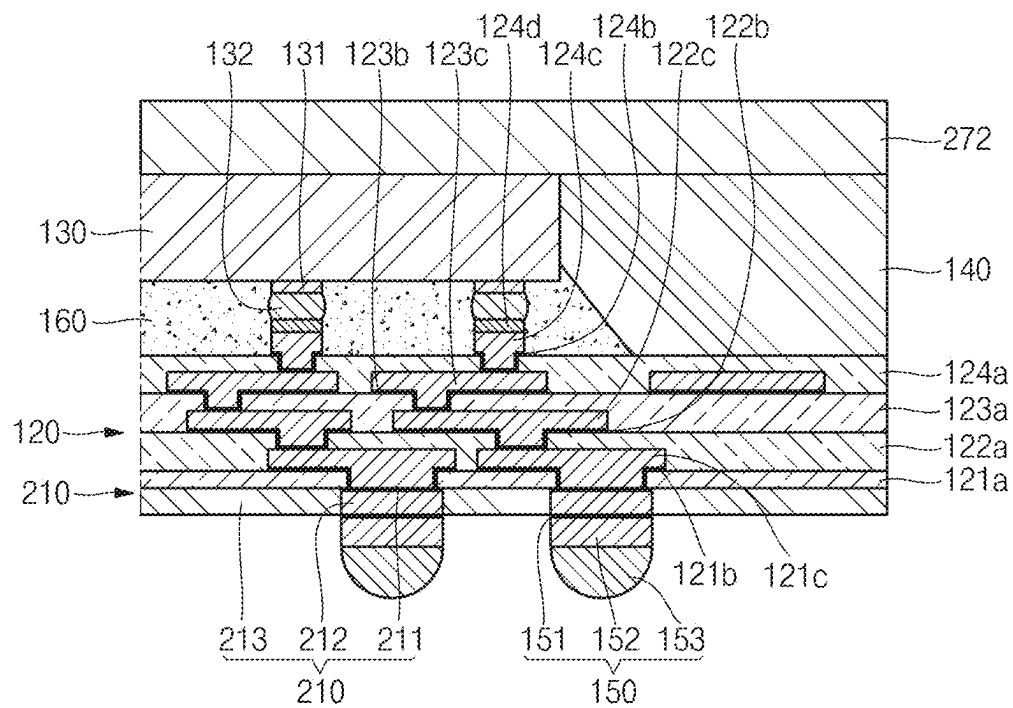


FIG. 4K

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DEVICE CHIP SCALE PACKAGE INCLUDING A PROTECTIVE LAYER AND METHOD OF MANUFACTURING A DEVICE CHIP SCALE PACKAGE

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. application Ser. No. 17/720,211 filed Apr. 13, 2022, which is a continuation of U.S. application Ser. No. 16/387,924 filed Apr. 18, 2019 (abandoned). Said application Ser. No. 17/720,211, said application Ser. No. 16/387,924, US Pub. No. US 2020/0335441 A1, and US Pub. No. 2022/0238441 A1 are hereby incorporated herein by reference in their entireties.

TECHNICAL FIELD

The present disclosure relates, in general, to electronic devices, and more particularly, to semiconductor devices and methods for manufacturing semiconductor devices.

BACKGROUND

Prior semiconductor packages and methods for forming semiconductor packages are inadequate, for example resulting in excess cost, decreased reliability, relatively low performance, or package sizes that are too large. Further limitations and disadvantages of conventional and traditional approaches will become apparent to one of skill in the art, through comparison of such approaches with the present disclosure and reference to the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows a cross-sectional view of an example semiconductor device.

FIGS. 2A to 2L show cross-sectional views of an example method for manufacturing an example semiconductor device.

FIG. 3 shows a cross-sectional view of another example semiconductor device.

FIGS. 4A to 4K show cross-sectional views of an example method for manufacturing another example semiconductor device.

The following discussion provides various examples of semiconductor devices and methods of manufacturing semiconductor devices. Such examples are non-limiting, and the scope of the appended claims should not be limited to the particular examples disclosed. In the following discussion, the terms “example” and “e.g.” are non-limiting.

The figures illustrate the general manner of construction, and descriptions and details of well-known features and techniques may be omitted to avoid unnecessarily obscuring the present disclosure. In addition, elements in the drawing figures are not necessarily drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help improve understanding of the examples discussed in the present disclosure. The same reference numerals in different figures denote the same elements.

The terms “or” and “and/or” include any single item, or any combination of the items, in the list joined by “or” or “and/or”. As used in this disclosure, the singular forms are intended to include the plural forms as well, unless the context clearly indicates otherwise.

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The terms “comprises,” “comprising,” “includes,” and/or “including,” are “open ended” terms and specify the presence of stated features, but do not preclude the presence or addition of one or more other features.

The terms “first,” “second,” etc. may be used herein to describe various elements, and these elements should not be limited by these terms. These terms are only used to distinguish one element from another. Thus, for example, a first element discussed in this disclosure could be termed a second element without departing from the teachings of the present disclosure.

Unless specified otherwise, the term “coupled” may be used to describe two elements directly contacting each other or describe two elements indirectly connected by one or more other elements. For example, if element A is coupled to element B, then element A can be directly contacting element B or indirectly connected to element B by an intervening element C. Similarly, the terms “over” or “on” may be used to describe two elements directly contacting each other or describe two elements indirectly connected by one or more other elements.

DESCRIPTION

In one example, a semiconductor device comprises a redistribution layer (RDL) substrate having a top surface and a bottom surface, wherein the RDL substrate comprises a filler-free dielectric material, an electronic device on the top surface of the RDL substrate, an electrical interconnect on the bottom surface of the RDL substrate and electrically coupled to the electronic device, a first protective material contacting a side surface of the electronic device and the top surface of the RDL substrate, and a second protective material contacting a side surface of the electrical interconnect and the bottom surface of the RDL substrate.

In another example, a method to manufacture a semiconductor device comprises forming a base structure having a conductive post, forming a redistribution layer (RDL) substrate on the base structure, placing an electronic device on a top surface of the RDL substrate, and forming a protective material contacting a side surface of the electronic device and the top surface of the RDL substrate.

In a further example, a method to manufacture a semiconductor device comprises forming a redistribution layer (RDL) substrate on a first carrier, the RDL substrate having a top surface and a bottom surface, placing an electronic device on the top surface of the RDL substrate, forming a first protective material using a first molding operation, wherein the first protective material contacts a side surface of the electronic device and the top surface of the RDL substrate, attaching a second carrier to the first protective material, removing the first carrier from the RDL substrate, forming a conductive post on the bottom surface of the RDL substrate using a first plating operation, and forming a second protective material using a second molding operation, wherein the second protective material contacts a side surface of the conductive post and the bottom surface of the RDL substrate.

Other examples are included in the present disclosure. Such examples may be found in the figures, in the claims, and/or in the description of the present disclosure.

FIG. 1 shows a cross-sectional view of an example semiconductor device. In the example shown in FIG. 1, semiconductor device 100 can comprise a base structure 110, a substrate 120, an electronic device 130, an encapsulant 140 and interconnects 150. In addition, semiconductor device 100 can further comprise a dielectric layer 160

between substrate **120** and electronic device **130**. In some examples, electronic device **130** can comprise an active device such as a semiconductor die or transistor, and in other examples electronic device **130** can comprise a passive device such as a resistor, a capacitor, an inductor, a connector, or equivalents.

Base structure **110** can comprise a conductive layer **112** and a dielectric layer **113**. Substrate **120** can comprise dielectric layers **121a**, **122a**, **123a** and **124a** and conductive layers **121b**, **122b**, **123b**, **124b**, **121c**, **122c**, **123c**, **124c** and **124d**. Electronic device **130** can comprise interconnects **131** and **132**. Encapsulant **140** can contact a top surface of substrate **120** and a side surface of electronic device **130**. In addition, interconnects **150** can comprise conductive layers **151**, **152** and **153** and can be located on a bottom surface of base structure **110**.

Base structure **110**, substrate **120**, encapsulant **140** and interconnects **150** can be referred to as a semiconductor package **190** or a package **190**. In addition, semiconductor package **190** can protect electronic device **130** from external elements and/or environmental exposure. In addition, semiconductor package **190** can provide electrical coupling between an external device (not shown) and electronic device **130**.

FIGS. 2A to 2L show cross-sectional views of an example method for manufacturing an example semiconductor device. FIG. 2A shows a process of providing a carrier **171** at an early stage of manufacture.

In the example shown in FIG. 2A, carrier **171** is substantially planar. In some examples, carrier **171** can be referred to as a board, a wafer, a panel or a strip as well. In addition, in some examples, carrier **171** can be made of any one or more of a metal (e.g., SUS), a wafer (e.g., silicon), ceramic (e.g., alumina), glass (e.g., soda-lime glass), or any equivalent. Carrier **171** can have a thickness in the range from approximately 500 μm to approximately 1500 μm and a width in the range from approximately 100 mm to approximately 500 mm. Carrier **171** can function to handle multiple components in an integrated manner during processes of forming base structure **110**, forming substrate **120**, and attaching and encapsulating electronic device **130**. In some examples, carrier **171** can be commonly applied to all examples of this disclosure.

FIG. 2B shows a process of forming conductive layers **111** and **112** at a later stage of manufacture. In the example shown in FIG. 2B, conductive layer **111** can be formed on carrier **171**. In some examples, conductive layer **111** can be referred to as a seed layer or a base layer. In some examples, seed layer **111** can be made of any of a variety of electrically conductive materials (e.g., titanium, tungsten, titanium/tungsten, copper, gold, silver, palladium, nickel, or an equivalent thereof). In addition, in some examples, seed layer **111** can be formed using any of a variety of processes (e.g., sputtering, electroless plating, electroplating, physical vapor deposition (PVD), chemical vapor deposition (CVD), metal organic chemical vapor deposition (MOCVD), atomic layer deposition (ALD), low pressure chemical vapor deposition (LPCVD), plasma enhanced chemical vapor deposition (PECVD), or an equivalent thereof). Seed layer **111** can have a thickness in the range from approximately 500 angstrom ($\text{\AA}$) to approximately 3000 $\text{\AA}$. Seed layer **111** can facilitate forming conductive layer **112** to a predetermined thickness at a later stage of manufacture.

In addition, in the example shown in FIG. 2B, conductive layer **112** that is relatively thick can be formed on seed layer **111** that is relatively thin. In some examples, a pattern can be formed on seed layer **111** using a patterned mask (not

shown) and relatively thick conductive layer **112** can be formed only within the pattern. In some examples, conductive layer **112** can be referred to as a conductive post or an under bump metal. In some examples, conductive post **112** can be made of any of a variety of electrically conductive materials (e.g., copper, gold, silver, or an equivalent thereof). Conductive post **112** can be formed using any of a variety of processes (e.g., sputtering, electroless plating, electroplating, PVD, CVD, MOCVD, ALD, LPCVD, PECVD or an equivalent thereof). After conductive post **112** is formed, the patterned mask can be removed. Additionally, relatively thin seed layer **111** formed around relatively thick conductive post **112** can also be removed using, for example, a soft etching process. Conductive post **112** can have a thickness in the range from approximately 1 μm to approximately 10 μm . Conductive post **112** can function to build up substrate **120** on conductive post **112** and to form interconnects **150** under conductive post **112** at later stages of manufacture.

FIG. 2C shows a process of forming dielectric layer **113** at a later stage of manufacture. In the example shown in FIG. 2C, conductive posts **111** and **112** formed on carrier **171** can be covered by dielectric layer **113**. In some examples, dielectric layer **113** can be formed using a molding operation, and dielectric layer **113** can contact a side of conductive post **112**. In some examples, dielectric layer **113** can cover top and side surfaces of conductive post **112**, and dielectric layer **113** may not cover a bottom surface of conductive post **112**. In some examples, dielectric layer **113** may not cover the top surface of conductive post **112** to allow the top surface of conductive post **112** to be exposed to the outside through dielectric layer **113**. In some examples, dielectric layer **113** can be referred to as an encapsulant, a sealant, an epoxy molding compound, a protective material, or an epoxy molding resin. In addition, in some examples, encapsulant **113** can also be referred to as an encapsulation part, a molding part, a protection part, or a body. In some examples, encapsulant **113** can comprise, but not limited to, an organic resin, an inorganic filler, a curing agent, a catalyst, a colorant, a flame retardant, or equivalents of the foregoing. Encapsulant **113** can be formed by any of a variety of processes including a molding operation. In some examples, encapsulant **113** can be formed by, but not limited to, compression molding, transfer molding, liquid-phase encapsulant molding, vacuum lamination, paste printing or film assist molding. Encapsulant **113** can have a thickness in the range from approximately 50 μm to approximately 100 μm . Encapsulant **113** can encapsulate conductive posts **111** and **112** to reduce or prevent substrate **120** from warping at a later stage.

FIG. 2D shows a process for removing portions of conductive post **112** and encapsulant **113** at a later stage of manufacture. In the example shown in FIG. 2D, conductive post **112** and a top surface of encapsulant **113** are subjected to removal, such as by grinding or etching to make conductive post **112** and the top surface of encapsulant **113** coplanar. In some examples, conductive post **112** and the top surface of encapsulant **113** can be made to be coplanar by grinding and/or etching to improve planarity of substrate **120** formed on conductive post **112** and encapsulant **113**. In such a manner, base structure **110** can be completed, substrate **120** can later be formed on base structure **110**, and interconnects **150** can be formed under base structure **110**.

FIG. 2E shows a process of forming substrate **120** at a later stage of manufacture. In the example shown in FIG. 2E, substantially planar substrate **120** can be directly formed or built up on base structure **110**. In an example, dielectric

layers **121a**, **122a**, **123a** and **124a** and conductive layers **121b**, **122b**, **123b**, **124b**, **121c**, **122c**, **123c**, **124c** and **124d** can be built up multiple times on base structure **110** to complete substrate **120**.

In some examples, dielectric layer **121a** can cover a top surface of base structure **110**. Since the top surface of base structure **110** can be planar, dielectric layer **121a** can also be planar. In some examples, dielectric layer **121a** can be referred to as a passivation layer, an insulation layer or a protection layer. Dielectric layer **121a** can be made of any of a variety of electrically non-conductive materials (e.g., Si_3N_4 , SiO_2 , SiON , polyimide (PI), benzocyclobutene (BCB), polybenzoxazole (PBO), bismaleimide triazine (BT), epoxy resin, phenol resin, silicone resin, acrylate polymer, or an equivalent thereof). In addition, dielectric layer **121a** can be formed using any of a variety of processes (e.g., PVD, CVD, MOCVD, ALD, LPCVD, PECVD, printing, spin coating, spray coating, sintering, thermal oxidation, or an equivalent thereof). In some examples, dielectric layer **121a** can be patterned to form an opening exposing conductive post **112** while covering encapsulant **113**. Dielectric layer **121a** can have a thickness in the range from approximately 1 μm to approximately 10 μm , and opening can have a width in the range from approximately 5 μm to approximately 70 μm .

In some examples, conductive layer **121b** can be conformally formed on dielectric layer **121a** and exposed conductive post **112**. In some examples, conductive layer **121b** can be referred to as a seed layer or base layer. In some examples, seed layer **121b** can be formed on a top surface of dielectric layer **121a**, a side wall of the opening, and the top surface of conductive post **112**.

In some examples, seed layer **121b** can be made of any of a variety of electrically conductive materials (e.g., titanium, tungsten, titanium/tungsten, copper, gold, silver, palladium, nickel, or an equivalent thereof). In addition, in some examples, seed layer **121b** can be formed using any of a variety of processes (e.g., sputtering, electroless plating, electroplating, PVD, CVD, MOCVD, ALD, LPCVD, PECVD, or an equivalent thereof). Seed layer **121b** can have a thickness in the range from approximately 500 Å to approximately 3000 Å. Seed layer **121b** can facilitate forming conductive layer **121c** to a predetermined thickness at a later stage of manufacture.

Although not shown, a mask can be formed on seed layer **121b** to then be patterned by a general photolithography process. In some examples, seed layer **121b** can be exposed to the outside by the patterned mask. In some examples, the patterned mask can include an opening that can expose a portion of seed layer **121b** to the outside. In some examples, the mask can be referred to as a photoresist or a resin.

In some examples, conductive layer **121c** that is relatively thick can be formed in the openings of the patterned mask on the exposed portions of seed layer **121b** that is relatively thin. Here, since a pattern has already been formed using the mask, relatively thick conductive layer **121c** can be formed only within the openings of the formed pattern. In some examples, conductive layer **121c** can be referred to as a redistribution layer (RDL), a wiring pattern or a circuit pattern. In some examples, redistribution layer **121c** can be made of any of a variety of electrically conductive materials (e.g., copper, gold, silver or an equivalent thereof). Redistribution layer **121c** can be formed using any of a variety of processes (e.g., sputtering, electroless plating, electroplating, PVD, CVD, MOCVD, ALD, LPCVD, PECVD, or an equivalent thereof). After redistribution layer **121c** is formed, the patterned mask can be removed. Additionally,

relatively thin seed layer **121b** formed under the patterned mask can be removed using, for example, a soft etching process after the patterned mask is removed. Redistribution layer **121c** can have a thickness in the range from approximately 2 μm to approximately 10 μm . Redistribution layer **121c** can function to electrically connect interconnects **131** and **132** of electronic device **130** to conductive post **112** of base structure **110**.

The aforementioned processes are repeated multiple times to form substrate **120** on base structure **110**. Here, conductive layer **124c** formed on the topmost surface of substrate **120** can be referred to as a conductive pad, a micro pad or a bond pad. In some examples, conductive pad **124c** can be formed to protrude a predetermined height from the top surface of substrate **120**. Conductive pad **124c** can have a width in the range from approximately 1 μm to approximately 80 μm .

In some examples, in order to prevent conductive pad **124c** from being oxidized, an antioxidant layer **124d** can be further formed on a top surface of conductive pad **124c**. In some examples, antioxidant layer **124d** can be referred to as a corrosion prevention layer or a solder spread improvement layer. In some examples, antioxidant layer **124d** can be made of tin, gold, silver, nickel, palladium or an equivalent thereof. Antioxidant layer **124d** can be formed using any of a variety of processes (e.g., sputtering, electroless plating, electroplating, PVD, CVD, MOCVD, ALD, LPCVD, PECVD, or an equivalent thereof). Antioxidant layer **124d** can have a width in the range from approximately 1 μm to approximately 80 μm .

In some examples, substrate **120** can be referred to as an interconnection structure, a build-up structure, a circuit stack structure, an RDL structure, or a printed circuit board. In the example shown in this disclosure, substrate **120**, which comprises four dielectric layers **121a**, **122a**, **123a** and **124a**, four conductive layers **121b**, **122b**, **123b** and **124b** and four conductive layers **121c**, **122c**, **123c** and **124c**, is illustrated. However, the quantity of these layers can be smaller than or greater than four.

Substrate **120** is presented as a redistribution layer (RDL) substrate in the example of FIG. 2. RDL substrates can comprise one or more conductive redistribution layers and one or more dielectric layers that (a) can be formed layer by layer over an electronic device to which the RDL substrate is to be electrically coupled, or (b) can be formed layer by layer over a carrier that can be entirely removed or at least partially removed after the electronic device and the RDL substrate are coupled together. RDL substrates can be manufactured layer by layer as a wafer-level substrate on a round wafer in a wafer-level process, and/or as a panel-level substrate on a rectangular or square panel carrier in a panel-level process. RDL substrates can be formed in an additive buildup process that can include one or more dielectric layers alternately stacked with one or more conductive layers that define respective conductive redistribution patterns or traces configured to collectively (a) fan-out electrical traces outside the footprint of the electronic device, and/or (b) fan-in electrical traces within the footprint of the electronic device. The conductive patterns can be formed using a plating process such as, for example, an electroplating process or an electroless plating process. The conductive patterns can comprise an electrically conductive material such as, for example, copper or other plateable metal. The locations of the conductive patterns can be made using a photo-patterning process such as, for example, a photolithography process and a photoresist material to form a photolithographic mask. The dielectric layers of the RDL

substrate can be patterned with a photo-patterning process, which can include a photolithographic mask through which light is exposed to photo-pattern desired features such as vias in the dielectric layers. Thus, the dielectric layers can be made from photo-definable organic dielectric materials such as, for example, polyimide (PI), benzocyclobutene (BCB), or polybenzoxazole (PBO). Such dielectric materials can be spun-on or otherwise coated in liquid form, rather than attached as a pre-formed film. To permit proper formation of desired photo-defined features, such photo-definable dielectric materials can omit structural reinforcements or can be filler-free, without strands, weaves, or other particles, that could interfere with the light from the photo-patterning process. In some examples, such filler-free characteristics of filler-free dielectric materials can permit a reduction of the thickness of the resulting dielectric layer. Although the photo-definable dielectric materials described above can be organic materials, in other examples the dielectric materials of the RDL substrates can comprise one or more inorganic dielectric layers. Some examples of inorganic dielectric layer(s) can comprise silicon nitride (Si_3N_4), silicon oxide (SiO_2), and/or SiON. The inorganic dielectric layer(s) can be formed by growing the inorganic dielectric layers using an oxidation or nitridization process instead using photo-defined organic dielectric materials. Such inorganic dielectric layers can be filler-free, without strands, weaves, or other dissimilar inorganic particles. In some examples, the RDL substrates can omit a permanent core structure or carrier such as, for example, a dielectric material comprising bismaleimide triazine (BT) or FR4 and these types of RDL substrates can be referred to as a coreless substrate.

In other examples, substrate **120** can be a pre-formed substrate. The pre-formed substrate can be manufactured prior to attachment to an electronic device and can comprise dielectric layers between respective conductive layers. The conductive layers can comprise copper and can be formed using an electroplating process. The dielectric layers can be relatively thicker non-photo-definable layers that can be attached as a pre-formed film rather than as a liquid and can include a resin with fillers such as strands, weaves, and/or other inorganic particles for rigidity and/or structural support. Since the dielectric layers are non-photo-definable, features such as vias or openings can be formed by using a drill or laser. In some examples, the dielectric layers can comprise a prepreg material or Ajinomoto Buildup Film (ABF). The pre-formed substrate can include a permanent core structure or carrier such as, for example, a dielectric material comprising bismaleimide triazine (BT) or FR4, and dielectric and conductive layers can be formed on the permanent core structure. In other examples, the pre-formed substrate can be a coreless substrate which omits the permanent core structure, and the dielectric and conductive layers can be formed on a sacrificial carrier that is removed after formation of the dielectric and conductive layers and before attachment to the electronic device. The pre-formed substrate can be referred to as a printed circuit board (PCB) or a laminate substrate. Such pre-formed substrate can be formed through a semi-additive or modified-semi-additive process.

FIG. 2F shows a process of attaching electronic device **130** at a later stage of manufacture. In the example shown in FIG. 2F, electronic device **130** can be electrically connected to substrate **120**. In some examples, a pick-and-place equipment (not shown) can pick up electronic device **130** to place electronic device **130** on conductive pad **124c** of substrate **120**. Next, electronic device **130** can be electrically con-

nected to substrate **120**, for example, by mass reflow, thermal compression or laser assist bonding.

In some examples, electronic device **130** can be referred to as a semiconductor die or a semiconductor chip. In addition, in some examples, electronic device **130** can comprise at least one of a logic die, a micro control unit, a memory, a digital signal processor, a network processor, a power management unit, an audio processor, an RF circuit, a wireless baseband system on chip processor, an application specific integrated circuit or an equivalent thereof.

In some examples, electronic device **130** can include an active region and a non-active region. In addition, in some examples, active region can be disposed to face substrate **120**. In addition, in some examples, active region can include interconnects **131**. In some examples, interconnects **131** can be referred to as die pads, bond pads, aluminum pads, conductive pillars or conductive posts. Interconnects **131** can have a width in the range from approximately $2\text{ }\mu\text{m}$ to approximately $80\text{ }\mu\text{m}$.

In addition, each of interconnects **131** can be connected to a conductive pad **124c** and/or antioxidant layer **124d** of substrate **120** through low melting point material **132**. In an example, low melting point material **132** can comprise any one or more of Sn, Ag, Pb, Cu, Sn—Pb, Sn37-Pb, Sn95-Pb, Sn—Pb—Ag, Sn—Cu, Sn—Ag, Sn—Au, Sn—Bi, Sn—Ag—Cu, or any equivalent. Interconnect **131** of electronic device **130** and conductive pad **124c** of substrate **120** can be electrically connected to each other by low melting point material **132**.

In some examples, dielectric layer **160** can be further filled between substrate **120** and electronic device **130**. In some examples, dielectric layer **160** can surround interconnects **131** of electronic device **130**, low melting point material **132**, conductive pad **124c** and antioxidant layer **124d**. In some examples, dielectric layer **160** can be referred to as an underfill, a capillary underfill (CUF), or a non-conductive paste. In some examples, underfill **160** can be a resin without an inorganic filler. In some examples, after electronic device **130** is electrically connected to substrate **120**, underfill **160** can be injected into gaps between electronic device **130** and substrate **120** by a capillary to then be cured. In some examples, underfill **160** can be formed around the perimeter of the gap between electronic device **130** and substrate **120**, and then underfill **160** will fill the gap through capillary forces. In some examples, underfill **160** can first be dispensed to cover conductive pad **124c** disposed on substrate **120**, and interconnect **131** of electronic device **130** and/or low melting point material **132** can then be electrically connected to conductive pad **124c** while passing through underfill **160**. Underfill **160** can prevent electronic device **130** from being electrically disconnected from substrate **120** due to physical shock or chemical shock.

FIG. 2G shows an encapsulating process at a later stage of manufacture. In the example shown in FIG. 2G, electronic device **130** can be encapsulated by encapsulant **140**. In some examples, encapsulant **140** can contact top and side surfaces of electronic device **130** and can contact underfill **160**. However, in some examples, encapsulant **140** may not contact a bottom surface of electronic device **130** and a bottom surface of underfill **160**. In some examples, encapsulant **140** may not contact the top surface of electronic device **130** to allow the top surface of electronic device **130** to be exposed to the outside through encapsulant **140**. In some examples, encapsulant **140** can be referred to as an epoxy molding compound, an epoxy molding resin, a protective material, or a sealant. In addition, in some examples, encapsulant **140** can be referred to as a molding part, a

sealing part, an encapsulation part, a package or a body. In some examples, encapsulant **140** can comprise, but not limited to, an organic resin, an inorganic filler, a curing agent, a catalyst, a colorant, or a flame retardant. Encapsulant **140** can be formed by any of a variety of processes. In some examples, encapsulant **140** can be formed by, but not limited to, a molding operation, compression molding, transfer molding, liquid-phase encapsulant molding, vacuum lamination, paste printing or film assist molding. Encapsulant **140** can have a thickness in the range from approximately 50 μm to approximately 1000 μm . Encapsulant **140** can encapsulate electronic device **130** to protect electronic device **130** from external elements and/or environmental exposure. In some examples, encapsulant **140** can serve as underfill, such as a molded underfill formed between substrate **120** and electronic device **130**.

In some examples, a material forming encapsulant **140** can be the same with or different from that of base structure **110**. When the material forming encapsulant **140** encapsulating electronic device **130** is the same with that of base structure **110**, the coefficient of thermal expansion (CTE) of upper and lower regions of semiconductor device **100** can be substantially the same with each other to suppress warpage of semiconductor device **100**.

For example, the CTE of substrate **120** can be different from that of encapsulant **140**. Therefore, substrate **120** and encapsulant **140** can tend to warp or bend in one direction by the heat applied during the manufacturing process of the semiconductor package or the heat generated during electrical operation of the semiconductor package. However, encapsulants **113** and **140** can be selected to have same or similar CTEs, and can be formed on opposite upper and lower portions of the substrate **120**, respectively. Thus, expansion or warpage due to the difference between the CTEs of encapsulant **140** and substrate **120** will tend to counteract expansion or warpage due to the difference between the CTEs of encapsulant **113** and substrate **120**. Accordingly, even if heat is applied during the manufacturing process of the semiconductor package or heat is generated during the electrical operation of the semiconductor package, the amount of warpage that the semiconductor package is bent in one direction can be suppressed or reduced. In some examples, the CTE of substrate **120** can be greater than the CTE of encapsulant **140** and greater than the CTE of encapsulant **113**.

There can also be examples where the material forming encapsulant **140** encapsulating electronic device **130** can be made different from that of encapsulant **113** and/or base structure **110** while still improving the warpage of semiconductor device **100**. For example, the material or CTEs of encapsulant **140** and of encapsulant **113** can be selected, even if different from each other, such that when also considering the thickness of encapsulant **140**, the thickness of encapsulant **113**, and/or the presence of electronic device **130**, the net effect is that warpage due to the interface between substrate **120** and encapsulant **140** counteracts warpage along the interface between substrate **120** and encapsulant **113**.

FIG. 2H shows a process of removing a portion of molding part **140** at a later stage of manufacture. In the example shown in FIG. 2H, molding part **140** can be subjected to grinding and/or etching, thereby exposing the top surface of electronic device **130** to the outside. The removing process can be performed until the thickness of electronic device **130** becomes smaller than approximately

500 μm . As the result of the removing process, a top surface of molding part **140** can be coplanar with the top surface of electronic device **130**.

FIG. 2I shows a process of attaching a carrier **172** at a later stage of manufacture. In the example shown in FIG. 2I, carrier **172** can be attached to molding part **140** and the top surface of electronic device **130**. In some examples, carrier **172** can be attached to molding part **140** and the top surface of electronic device **130** using a temporary adhesive layer. The temporary adhesive layer can be made of a material configured to lose its adhesiveness when exposed to heat or light. Upper carrier **172** can fix or support the device while removing lower carrier **171**. Upper carrier **172** can be substantially planar. In some examples, upper carrier **172** can be referred to as a board, a wafer, a panel or a strip as well. In addition, in some examples, upper carrier **172** can be made of any one or more of a metal (e.g., SUS), a wafer (e.g., silicon), ceramic (e.g., alumina), glass (e.g., soda-lime glass), or any equivalent. Upper carrier **172** can have a thickness in the range from approximately 500 μm to approximately 1500 μm and a width in the range from approximately 100 mm to approximately 500 mm.

FIG. 2J shows a process of removing carrier **171** at a later stage of manufacture. In the example shown in FIG. 2J, carrier **171** can be removed from base structure **110**. In some examples, carrier **171** can be removed by grinding and/or etching using a grinding operation and/or an etching operation. In some examples, when grinding and/or etching is performed on carrier **171**, seed layer **111** formed on the bottom surface of conductive post **112** can also be removed. Therefore, the bottom surface of conductive post **112** can be exposed to the outside through encapsulant **113**. In some examples, the bottom surface of conductive post **112** can be coplanar with the bottom surface of encapsulant **113**.

FIG. 2K shows a process of removing carrier **172** at a later stage of manufacture. In the example shown in FIG. 2K, upper carrier **172** can also be removed. As described above, the top surface of electronic device **130** and the top surface of encapsulant **140** can be coplanar on semiconductor device **100**, while the bottom surface of conductive post **112** of base structure **110** and the bottom surface of encapsulant **113** can be coplanar under semiconductor device **100**. In some examples, carrier **172** may be removed using grinding operation and/or an etching operation in the same manner or in a manner similar to the removal of carrier **171** as discussed with respect to FIG. 2J, above.

FIG. 2L shows a process of forming interconnects **150** at a later stage of manufacture. In the example shown in FIG. 2L, conductive layer **151** that is relatively thin can be formed on the entire bottom surface of base structure **110**, and conductive layer **152** that is relatively thick can be formed on relatively thin conductive layer **151**. In some examples, relatively thin conductive layer **151** can be referred to as a seed layer or base layer. In some examples, seed layer **151** can be formed on bottom surfaces of conductive post **112** and encapsulant **113**.

Seed layer **151** can be made of any of a variety of electrically conductive materials (e.g., titanium, tungsten, titanium/tungsten, copper, gold, silver, palladium, nickel or an equivalent thereof). In addition, in some examples, seed layer **151** can be formed using any of a variety of processes (e.g., sputtering, electroless plating, electroplating, PVD, CVD, MOCVD, ALD, LPCVD, PECVD, or an equivalent thereof). Seed layer **151** can have a thickness in the range from approximately 500 $\text{\AA}$ to approximately 3000 $\text{\AA}$. Seed layer **151** can facilitate forming conductive layer **152** to a predetermined thickness at a later stage of manufacture.

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In some examples, relatively thick conductive layer 152 can be formed on relatively thin seed layer 151. In some examples, a pattern or opening can be formed on seed layer 151 using a patterned mask (not shown) and relatively thick conductive layer 152 can be formed only within the pattern or the opening. In some examples, conductive layer 152 that is relatively thick can be formed in the patterns of the patterned mask on the exposed portions of seed layer 151 that is relatively thin. Here, since a pattern has already been formed using the mask, relatively thick conductive layer 152 can be formed only within the openings of the formed pattern. In some examples, conductive layer 152 can be referred to as a conductive pillar or a conductive post. In some examples, conductive pillar 152 can be made of any of a variety of electrically conductive materials (e.g., copper, gold, silver or an equivalent thereof). Conductive pillar 152 can be formed using any of a variety of processes (e.g., sputtering, electroless plating, electroplating, PVD, CVD, MOCVD, ALD, LPCVD, PECVD, or an equivalent thereof). After conductive pillar 152 is formed, the patterned mask can be removed. Additionally, relatively thin seed layer 151 formed around relatively thick conductive pillar 152 can also be removed using, for example, a soft etching process. Conductive pillar 152 can have a thickness in the range from approximately 5 μm to approximately 50 μm .

In some examples, interconnect tip 153 having a relatively low melting point material can be connected to conductive pillar 152. In some examples, interconnect tip 153 can have a lower melting point than conductive pillar 152. In some examples, interconnect tip 153 can be referred to as a solder ball, a solder bump, a solder cap, a conductive ball, a conductive bump, or a conductive cap. In some examples, after dispensing solder to a bottom surface of conductive pillar 152, interconnect tip 153 can be formed on the bottom surface of conductive pillar 152 by a mass reflow process. In some examples, the patterned mask that is used to form conductive pillar 152 can be re-used to form interconnect tip 153. In some examples, interconnect tip 153 can be formed in the patterns or openings of the patterned mask on the exposed portions of conductive pillar 152. Here, since a pattern has already been formed using the mask, interconnect tip 153 can be formed only within the openings of the pattern. In some examples, interconnect tip 153 can comprise any one or more of Sn, Ag, Pb, Cu, Sn—Pb, Sn37—Pb, Sn95—Pb, Sn—Pb—Ag, Sn—Cu, Sn—Ag, Sn—Au, Sn—Bi, Sn—Ag—Cu, or any equivalent. Interconnect tip 153 can have a thickness in the range from approximately 0.5 μm to approximately 30 μm and a width in the range from approximately 2 μm to approximately 80 μm . After interconnect tip 153 is formed, the patterned mask can be removed. In some examples, if interconnect tip 153 is formed using the patterned mask, seed layer 151 formed around conductive pillar 152 and interconnect tip 153 can now be removed using, but not limited to, a soft etching process.

As described above, interconnects 150, which comprises seed layer 151, conductive pillar 152 and interconnect tip 153, can be completed. Interconnects 150 can function to electrically connect semiconductor device 100 or semiconductor package 190 to an external device (not shown). Although interconnects 150 are shown as being formed after carrier 172 is removed, this is not a limitation of the present disclosure. In other examples, interconnects 150 can be formed before carrier 172 is removed.

FIG. 3 shows a cross-sectional view of another example semiconductor device. Semiconductor device 200 shown in FIG. 3 can have a different structure from that of semicon-

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ductor device 100 shown in FIG. 1 due to processing differences of manufacture. In the example shown in FIG. 3, semiconductor device 200 can comprise a substrate 120, an electronic device 130, an encapsulant 140, a base structure 210 and interconnects 150.

Substrate 120 can comprise dielectric layers 121a, 122a, 123a and 124a and conductive layers 121b, 122b, 123b, 124b, 121c, 122c, 123c, 124c and 124d. Electronic device 130 can comprise interconnects 131 and 132. Encapsulant 140 can contact a top surface of substrate 120 and a side surface of electronic device 130. Base structure 210 can comprise conductive layers 211 and 212 and a dielectric layer 213. In addition, interconnects 150 can be located on a bottom surface of base structure 210.

Substrate 120, encapsulant 140, base structure 210 and interconnects 150 can be referred to as a semiconductor package 290 or a package 290. Semiconductor package 290 can protect electronic device 130 from external elements and/or environmental exposure. In addition, semiconductor package 290 can provide electrical coupling between an external device (not shown) and electronic device 130.

FIGS. 4A to 4K show cross-sectional views of an example method for manufacturing another example semiconductor device. FIG. 4A shows a process of providing a carrier 271 at an early stage of manufacture.

In the example shown in FIG. 4A, carrier 271 can have substantially the same shape and characteristic with those of carrier 171 shown in FIG. 2A.

FIG. 4B shows a process of forming substrate 120 at a later stage of manufacture. In the example shown in FIG. 4B, substantially planar substrate 120 can be directly formed or built up on carrier 271. In an example, dielectric layers 121a, 122a, 123a and 124a and conductive layers 121b, 122b, 123b, 124b, 121c, 122c, 123c, 124c and 124d can be built up sequentially upon each other on carrier 271, thereby completing substrate 120.

In some examples, dielectric layer 121a can cover a top surface of carrier 271. Since the top surface of carrier 271 is formed to be planar, dielectric layer 121a can also be formed to be planar. In some examples, dielectric layer 121a can be referred to as a passivation layer, an insulation layer or a protection layer. Dielectric layer 121a can be made of any of a variety of electrically non-conductive materials (e.g., Si₃N₄, SiO₂, SiON, polyimide (PI), benzocyclobutene (BCB), polybenzoxazole (PBO), bismaleimide triazine (BT), epoxy resin, phenol resin, silicone resin, acrylate polymer, or an equivalent thereof). In addition, dielectric layer 121a can be formed using any of a variety of processes (e.g., PVD, CVD, MOCVD, ALD, LPCVD, PECVD, printing, spin coating, spray coating, sintering, thermal oxidation, or an equivalent thereof). In some examples, dielectric layer 121a can be patterned to form an opening exposing a portion of carrier 271. Dielectric layer 121a can have a thickness in the range from approximately 1 μm to approximately 10 μm and opening can have a width in the range from approximately 5 μm to approximately 70 μm .

In some examples, conductive layer 121b can be entirely formed on dielectric layer 121a and exposed regions of carrier 271. In some examples, conductive layer 121b can be referred to as a seed layer or base layer. In some examples, seed layer 121b can be formed on a top surface of dielectric layer 121a, a side wall of the opening, and a top surface of carrier 271, respectively, and all of these conductive layers 121b can be electrically connected to each other.

In some examples, seed layer 121b can be made of any of a variety of electrically conductive materials (e.g., titanium, tungsten, titanium/tungsten, copper, gold, silver, palladium,

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nickel or an equivalent thereof). In addition, in some examples, seed layer **121b** can be formed using any of a variety of processes (e.g., sputtering, electroless plating, electroplating, PVD, CVD, MOCVD, ALD, LPCVD, PECVD, or an equivalent thereof). Seed layer **121b** can have a thickness in the range from approximately 500 Å to approximately 3000 Å. Seed layer **121b** can facilitate forming conductive layer **121c** to a predetermined thickness at a later stage of manufacture.

Although not shown, a mask can be formed on seed layer **121b** to then be patterned by a general photolithographic etching process. In some examples, seed layer **121b** can be exposed to the outside by the patterned mask. In some examples, the mask can be referred to as a photoresist or a resin.

In some examples, conductive layer **121c** that is relatively thick can be formed in the openings of the patterned mask on the exposed portions of seed layer **121b** that is relatively thin. Here, since a pattern has already been formed using the mask, relatively thick conductive layer **121c** can be formed only within the openings of the pattern. In some examples, conductive layer **121c** can be referred to as a redistribution layer (RDL), a wiring pattern or a circuit pattern. In some examples, redistribution layer **121c** can be made of any of a variety of electrically conductive materials (e.g., copper, gold, silver or an equivalent thereof). Redistribution layer **121c** can be formed using any of a variety of processes (e.g., sputtering, electroless plating, electroplating, PVD, CVD, MOCVD, ALD, LPCVD, PECVD, or an equivalent thereof). After redistribution layer **121c** is formed, the patterned mask can be removed. Additionally, relatively thin seed layer **121b** formed under the patterned mask can be removed using a soft etching process after the patterned mask is removed. Redistribution layer **121c** can have a thickness in the range from approximately 2 μm to approximately 10 μm. Redistribution layer **121c** can function to electrically connect interconnects **131** and **132** of electronic device **130** to conductive post **212** of base structure **210**.

The aforementioned processes can be repeated multiple times, thereby completing substrate **120** on carrier **271**. Here, conductive layer **124c** formed on the topmost surface of substrate **120** can be referred to as a conductive pad, a micro pad or a bond pad. In some examples, conductive pad **124c** can be formed to protrude a predetermined height from the top surface of substrate **120**. Conductive pad **124c** can have a width in the range from approximately 2 μm to approximately 80 μm.

In some examples, in order to prevent conductive pad **124c** from being oxidized, an antioxidant layer **124d** can be further formed on a top surface of conductive pad **124c**. In some examples, antioxidant layer **124d** can be made of tin, gold, silver, nickel, palladium or an equivalent thereof. Antioxidant layer **124d** can be referred to as a corrosion prevention layer or a solder spread improvement layer. Antioxidant layer **124d** can be formed using any of a variety of processes (e.g., sputtering, electroless plating, electroplating, PVD, CVD, MOCVD, ALD, LPCVD, PECVD, or an equivalent thereof). Antioxidant layer **124d** can have a width in the range from approximately 1 μm to approximately 80 μm.

In some examples, substrate **120** can be referred to as an interconnection structure, a build-up structure, a circuit stack structure, a RDL structure, or a printed circuit board. In the example showing this disclosure, substrate **120**, which comprises four dielectric layers **121a**, **122a**, **123a** and **124a**, four conductive layers **121b**, **122b**, **123b** and **124b** and four

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conductive layers **121c**, **122c**, **123c** and **124c**, is illustrated. However, the quantity of these layers can be smaller than or greater than four.

FIG. 4C shows a process of attaching electronic device **130** at a later stage of manufacture. In the example shown in FIG. 4C, the process of attaching electronic device **130** can be similar to that of attaching electronic device **130** shown in FIG. 2F.

FIG. 4D shows an encapsulating process at a later stage of manufacture. In the example shown in FIG. 4D, the encapsulating process can be the same as or similar to that of FIG. 2G.

FIG. 4E shows a process of removing a portion of molding part **140** at a later stage of manufacture. In the example shown in FIG. 4E, the removing process can be the same as or similar to that of in FIG. 2H.

FIG. 4F shows a process of attaching carrier **272** at a later stage of manufacture. The process of attaching carrier **272** shown in FIG. 4F can be the same as or similar to that of attaching carrier **272** in FIG. 2I.

FIG. 4G shows a process of removing carrier **271** at a later stage of manufacture. In the example shown in FIG. 4G, carrier **271** can be removed from substrate **120**. In some examples, carrier **271** can be removed by grinding and/or etching. In some examples, when grinding and/or etching is performed on carrier **271**, seed layer **121b** formed on the bottom surface of substrate **120** can be removed. In some examples, the bottom surface of redistribution layer **121c** can be removed. Therefore, the bottom surface of redistribution layer **121c** of substrate **120** can be exposed to the outside through dielectric layer **121a**. In some examples, the bottom surface of redistribution layer **121c** can be coplanar with the bottom surface of dielectric layer **121a**.

FIG. 4H shows a process of forming conductive layers **211** and **212** at a later stage of manufacture. In the example shown in FIG. 4H, conductive layers **211** and **212** can be formed on the bottom surface of substrate **120**. In some examples, conductive layers **211** and **212** can be formed on dielectric layer **121a** of substrate **120** and the bottom surface of redistribution layer **121c**. In some examples, conductive layer **211** can be referred to as a seed layer or a base layer. In some examples, seed layer **211** can be made of any of a variety of electrically conductive materials (e.g., titanium, tungsten, titanium/tungsten, copper, gold, silver, palladium, nickel, or an equivalent thereof). In addition, in some examples, seed layer **211** can be formed using any of a variety of processes (e.g., sputtering, electroless plating, electroplating, PVD, CVD, MOCVD, ALD, LPCVD, PECVD, or an equivalent thereof). Seed layer **211** can have a thickness in the range from approximately 500 Å to approximately 3000 Å. Seed layer **211** can facilitate forming conductive layer **212** to a predetermined thickness at a later stage of manufacture.

In addition, in the example shown in FIG. 4H, conductive layer **212** that is relatively thick can be formed on seed layer **211** that is relatively thin. In some examples, a pattern or opening can be formed on seed layer **211** using a patterned mask and conductive layer **212** can be formed only within the pattern or opening. In some examples, conductive layer **212** can be formed in the openings of the patterned mask on the exposed portions of seed layer **211**. Here, since a pattern has already been formed using the mask, conductive layer **212** can be formed only within the openings of the formed pattern. In some examples, conductive layer **212** can be referred to as a conductive post or an under bump metal. In some examples, conductive post **212** can be made of any of a variety of electrically conductive materials (e.g., copper,

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gold, silver, or an equivalent thereof). Conductive post **212** can be formed using any of a variety of processes (e.g., sputtering, electroless plating, electroplating, PVD, CVD, MOCVD, ALD, LPCVD, PECVD, or an equivalent thereof). After conductive post **212** is formed, the patterned mask can be removed. Additionally, relatively thin seed layer formed around relatively thick conductive post **212** can also be removed using a soft etching process. Conductive post **212** can have a thickness in the range from approximately 1 μm to approximately 10 μm . Conductive post **212** can be electrically connected to interconnect **150** to be formed under substrate **120** and/or base structure **210** at later stages of manufacture.

FIG. **4I** shows a process of forming dielectric layer **213** at a later stage of manufacture. In the example shown in FIG. **4I**, conductive post **212** formed under substrate **120** can be covered by dielectric layer **213**. In some examples, dielectric layer **213** can cover bottom and side surfaces of conductive post **212**. Dielectric layer, however, **213** may not cover a top surface of conductive post **212**. In some examples, dielectric layer **213** may not cover the bottom surface of conductive post **212**, thereby allowing the bottom surface of conductive post **212** to be exposed to the outside through dielectric layer **213**. In some examples, dielectric layer **213** can be referred to as an encapsulant, a sealant, an epoxy molding compound or an epoxy molding resin. In addition, in some examples, encapsulant **213** can be referred to as an encapsulation part, a molding part, a protection part, or a body. In some examples, encapsulant **213** can comprise, but not limited to, an organic resin, an inorganic filler, a curing agent, a catalyst, a colorant, a flame retardant, or equivalents of the foregoing. Encapsulant **213** can be formed by any of a variety of processes. In some examples, encapsulant **213** can be formed by, but not limited to, compression molding, transfer molding, liquid-phase encapsulant molding, vacuum lamination, paste printing or film assist molding. Encapsulant **213** can have a thickness in the range from approximately 1 μm to approximately 10 μm . Encapsulant **213** can firmly encapsulate conductive post **212** to reduce or prevent substrate **120** from warping at a later stage.

FIG. **4J** shows a removing process at a later stage of manufacture. In the example shown in FIG. **4J**, conductive post **212** and a bottom surface of encapsulant **213** are subjected to grinding or etching to expose the bottom surface of conductive post **212** to the outside through the bottom surface of encapsulant **213**. In some examples, the bottom surface of conductive post **212** and the bottom surface of encapsulant **213** can be formed to be coplanar. As described above, base structure **210** can be completed and interconnects **150** can be formed under base structure **210** at a later stage of manufacture.

FIG. **4K** shows a process of forming interconnects **150** at a later stage of manufacture. In the example shown in FIG. **4K**, the process of forming interconnects **150** can be substantially the same with the process of forming interconnects **150** shown in FIG. **2K**.

Meanwhile, carrier **272** can be removed. As described above, the top surface of electronic device **130** can be coplanar with a top surface of encapsulant **140** on semiconductor device **200**.

In summary, a semiconductor device comprises a redistribution layer (RDL) substrate having a top surface and a bottom surface, wherein the RDL substrate comprises a filler-free dielectric material, an electronic device on the top surface of the RDL substrate, an electrical interconnect on the bottom surface of the RDL substrate and electrically coupled to the electronic device, a first protective material

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contacting a side surface of the electronic device and the top surface of the RDL substrate, and a second protective material contacting a side surface of the electrical interconnect and the bottom surface of the RDL substrate.

A method to manufacture a semiconductor device comprises forming a base structure having a conductive post, forming a redistribution layer (RDL) substrate on the base structure, placing an electronic device on a top surface of the RDL substrate, and forming a protective material contacting a side surface of the electronic device and the top surface of the RDL substrate.

An alternative method to manufacture a semiconductor device comprises forming a redistribution layer (RDL) substrate on a first carrier, the RDL substrate having a top surface and a bottom surface, placing an electronic device on the top surface of the RDL substrate, forming a first protective material using a first molding operation, wherein the first protective material contacts a side surface of the electronic device and the top surface of the RDL substrate, attaching a second carrier to the first protective material, removing the first carrier from the RDL substrate, forming a conductive post on the bottom surface of the RDL substrate using a first plating operation, and forming a second protective material using a second molding operation, wherein the second protective material contacts a side surface of the conductive post and the bottom surface of the RDL substrate.

The present disclosure includes reference to certain examples. It will be understood, however, by those skilled in the art that various changes may be made, and equivalents may be substituted, without departing from the scope of the disclosure. In addition, modifications may be made to the disclosed examples without departing from the scope of the present disclosure. Therefore, it is intended that the present disclosure not be limited to the examples disclosed, but that the disclosure will include all examples falling within the scope of the appended claims.

The invention claimed is:

1. A method to manufacture a semiconductor device, comprising:

providing a redistribution layer (RDL) substrate comprising a top side, a bottom side opposite the top side, and a conductive layer, wherein the conductive layer includes a plurality of pads, the plurality of pads including a first portion below the top side of the RDL substrate and a second portion protruding above the top side of the RDL substrate;

disposing an electronic device on the plurality of pads; providing a first protective material over a lateral side of the electronic device and the top side of the RDL substrate;

providing a conductive post on the bottom side of the RDL substrate using a first plating operation;

providing a base structure on the bottom side of the RDL substrate, the base structure comprising a second protective material covering a lateral side of the conductive post and the bottom side of the RDL substrate;

providing a base layer on a bottom side of the conductive post, wherein a lateral side of the base layer is exposed; and

providing an external interconnect on a bottommost side of the base layer and coupled with the conductive post, wherein the external interconnect is coupled with the conductive post through an opening in the base layer.

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2. The method of claim 1, wherein providing the external interconnect on the base layer comprises:

providing a pillar on the base layer using a second plating operation; and
providing a solder cap on the pillar.

3. The method of claim 1, wherein the base layer comprises a conductive material.

4. The method of claim 1, wherein:

the RDL substrate is provided on a first carrier; and further comprising:

attaching a second carrier to the first protective material; and

removing the first carrier from the RDL substrate to expose the bottom side of the RDL substrate and the conductive layer.

5. The method of claim 4, wherein the first carrier is removed using a grinding or etching operation.

6. The method of claim 4, wherein providing the external interconnect on the base layer comprises:

providing a pillar on the base layer using a second plating operation; and

providing a solder cap on the pillar; and further comprising removing the second carrier after the solder cap is provided.

7. The method of claim 1, wherein:

the second protective material is provided in a first molding operation; and

the first protective material is provided in a second molding operation.

8. A method to manufacture a semiconductor device, comprising:

providing a redistribution layer (RDL) substrate having a top side and a bottom side, wherein the RDL substrate comprises a filler-free dielectric material, and wherein the RDL substrate comprises a conductive layer exposed from the bottom side, and wherein the conductive layer includes a plurality of pads, the plurality of pads including a first portion below the top side of the RDL substrate and a second portion protruding above the top side of the RDL substrate;

providing an electronic device on the plurality of pads; providing a conductive post on the bottom side of the RDL substrate and electrically coupled to the electronic device;

providing a first protective material over a lateral side of the electronic device and the top side of the RDL substrate;

providing a base structure on the bottom side of the RDL substrate, the base structure comprising a second protective material on a lateral side of the conductive post and the bottom side of the RDL substrate; and

providing an external interconnect below a bottom side of the base structure to couple with the conductive post, wherein the external interconnect is coupled with the conductive post through an opening in the base structure, wherein the external interconnect directly contacts the conductive post.

9. The method of claim 8, wherein the external interconnect comprises a conductive pillar and an interconnect tip on the conductive pillar.

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10. The method of claim 8, wherein the external interconnect comprises a conductive pillar and a conductive bump on the conductive pillar.

11. The method of claim 8, wherein the external interconnect comprises a solder ball.

12. The method of claim 8, wherein the RDL substrate comprises a dielectric layer.

13. The method of claim 8, wherein the first protective material and the second protective material comprise a same material having a same coefficient of thermal expansion (CTE).

14. The method of claim 8, wherein the first protective material and the second protective material comprise different materials having a same coefficient of thermal expansion (CTE).

15. The method of claim 8, wherein the first protective material has a first coefficient of thermal expansion (CTE) and a first thickness, and the second protective material has a second CTE and a second thickness such that warpage between the first protective material and the RDL substrate counters warpage between the second protective material and the RDL substrate.

16. The method of claim 8, wherein:

the second protective material is provided in a first molding operation; and

the first protective material is provided in a second molding operation.

17. A semiconductor device, comprising:

a redistribution layer (RDL) substrate comprising a top side, a bottom side opposite the top side, and a conductive layer, wherein the conductive layer includes a plurality of pads, the plurality of pads including a first portion below the top side of the RDL substrate and a second portion protruding above the top side of the RDL substrate;

an electronic device on the plurality of pads;

a first protective material over a lateral side of the electronic device and the top side of the RDL substrate;

a conductive post on the bottom side of the RDL substrate using a first plating operation;

a base structure on the bottom side of the RDL substrate, the base structure comprising a second protective material covering a lateral side of the conductive post and the bottom side of the RDL substrate;

a base layer on a bottom side of the conductive post, wherein a lateral side of the base layer is exposed; and an external interconnect on a bottommost side of the base layer and coupled with the conductive post, wherein the external interconnect is coupled with the conductive post through an opening in the base layer.

18. The semiconductor device of claim 17, wherein the external interconnect on the base layer comprises:

a pillar on the base layer; and

a solder cap on the pillar.

19. The semiconductor device of claim 18, wherein the pillar extends beyond a bottom side of the second protective material.

20. The semiconductor device of claim 17, wherein the base layer comprises a conductive material.

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